

Amazon Product Analysis

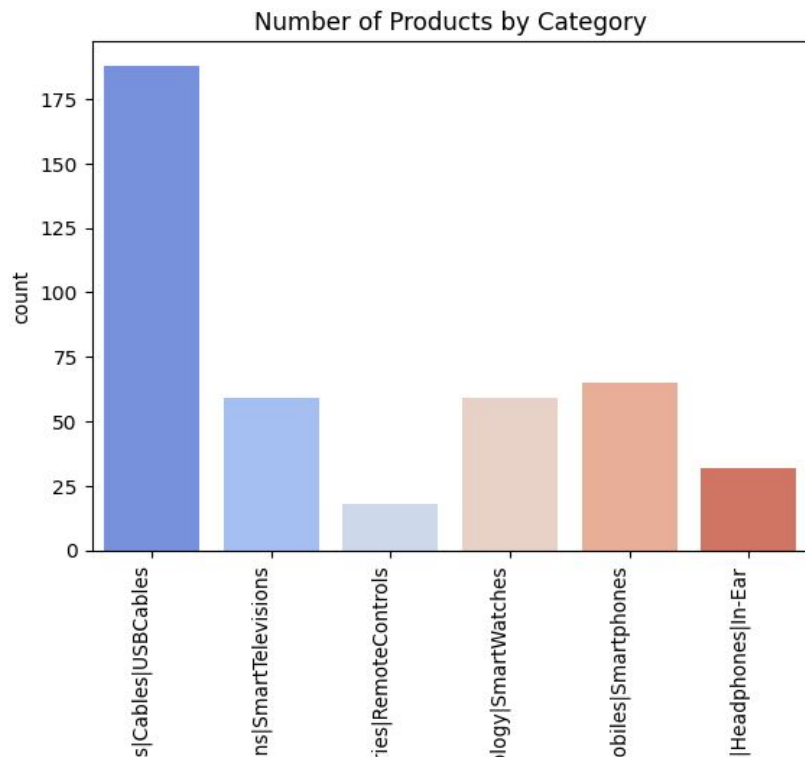
Initial Results and
Coding

Research Question 1

How does the price, rating and discount compare for products belonging to different categories?



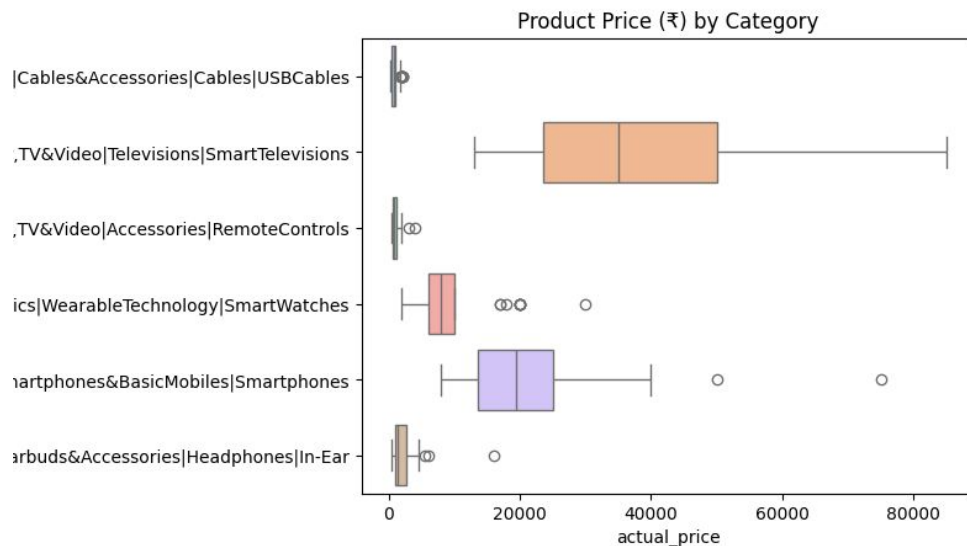
Product Categories



Top 3 products

1. USB cables
2. Smartphones
3. Smart watches

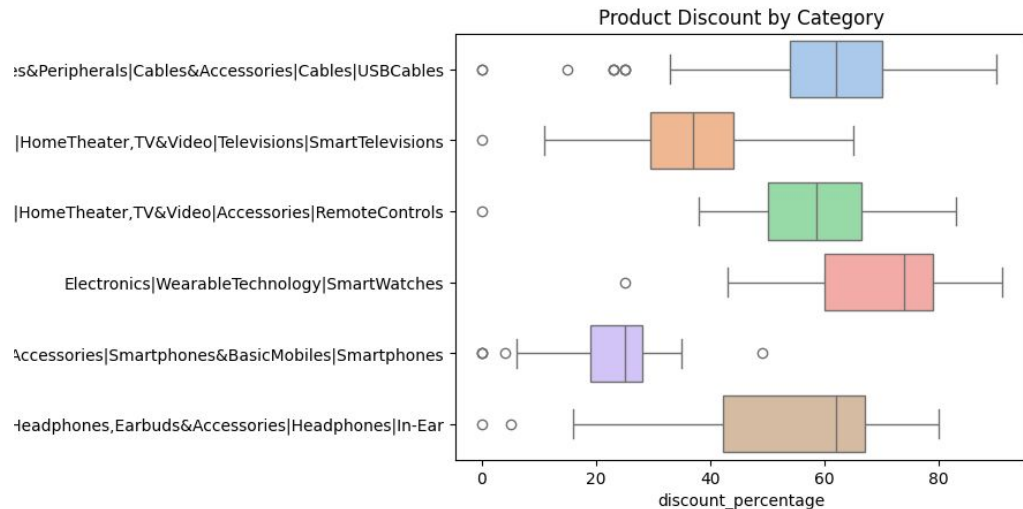
Product Price



Most expensive products

1. Smart televisions
2. Smartphones
3. Smart watches

Product Discount



Highest offered discount

1. Smart watches
2. USB cables
3. Remote controls

Key insight

Smart watches are one of the most expensive products with the highest offered discount

Research Question 2

How can the product characteristics be used to predict the product category?



Classifying Product Category

	Accuracy	Precision	Recall
Multinomial Logistic Regression	80%	80%	80%
KNN	76%	78%	76%
Decision Tree	89%	90%	89%

Model evaluation

1. **Accuracy:** 89% of the time the model correctly predicts the product category
2. **Precision:** 90% of all predictions correspond to the true product category
3. **Recall:** 89% of all products belonging to a category are identified

Key insight

Decision tree minimizes
misclassifications of
product category

Research Question 3

How can product descriptions be used to identify similar products?



Searching for Similar Products

ID	Category	Description	Cosine Similarity
Doc319	Smart watches	1.75" TruView™ display : See the clear, bigger picture...	0.69
Doc410	Smart watches	1.55 Always-on AMOLED Display with high 314 PPI Resolution...	0.5

Search query

1. **Product ID:** Doc401
2. **Product category:** smart watches
3. **Product description:** 1.75" TruView™ display: See the clear, bigger picture on the 1.75" touch screen...

Key insight

Top product matches all
belong to the same
category

Next Steps

- Further examine relationships between product characteristics
- Refine baseline classification models to improve category prediction performance
- Evaluate search relevance for all categories using product descriptions

